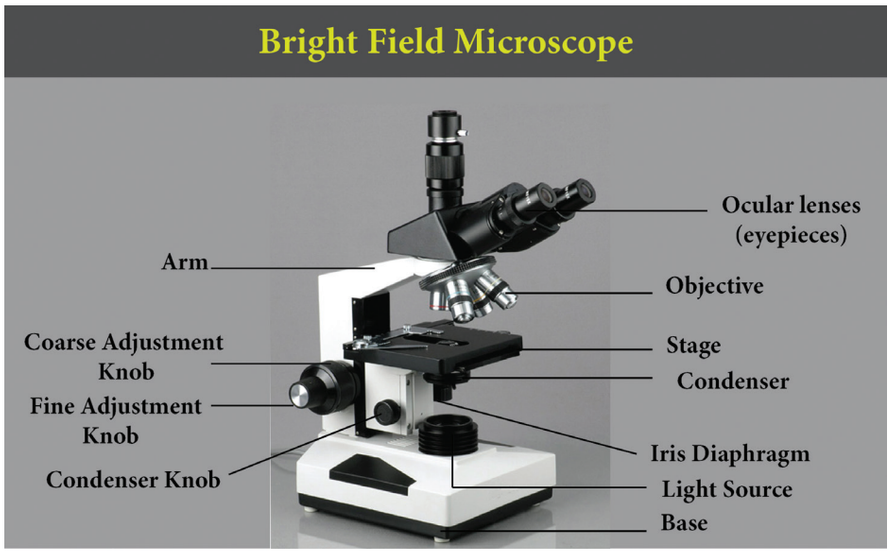


EDPBLH

Figure 2.6: Bright field Microscope

- There are two focusing knobs, the fine and the coarse adjustment knobs which are located on the arm. These are used to move either the stage or the nosepiece to focus the image.
- Mechanical stage is positioned about half way up the arm, which allows precise contact on moving the slide.
- The substage condenser is mounted within or beneath the stage and focuses a cone of light on the slide. In the simpler microscope, its position is fixed where as in advanced microscope it can be adjusted vertically.

The upper part of microscope arm holds the body assembly. The nose piece and one or more eyepieces or oculars are attached to it. The body assembly contains series of mirrors and prisms so that the barrel holding the eyepiece may be tilted for viewing. Three or five objectives with different magnification power are fixed to the nosepiece and can be rotated to the position beneath the body assembly. In bright field microscopy; the specimen is viewed against a bright background. The details of the image are defined by the surrounding light. A series of finely ground lenses forms an image which is many times larger than the real image. This magnification occurs when light rays from an illuminator (light source), pass through a condenser which has lenses that direct the light rays through the specimen. The light rays then pass into objective lens (the lens closest to the specimen). The image is again magnified by the ocular lens or the eyepiece. (Figure 2.7).

- Magnification is the process of enlarging the image of the specimen and can be calculated by multiplying the objective lens magnification power by ocular lens magnification power.

  Representative magnification values for a 10X ocular are:

  Scanning objective (4X) × (10X) = 40X magnification

  Low power objective (10X) × (10X) = 100X magnification

  High dry objective (40X) × (10X) = 400X magnification

  Oil immersion objective (100X) × (10X) = 1000X magnification

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